Patchwise Selective Semi-Amortized Physics Inference for Spatially Heterogeneous Lesion Phenotyping

Patch Study Draft v1

March 17, 2026

Abstract

This manuscript defines a **new patch-level study** that is distinct from the existing full-leaf selective semi-amortized paper. The new question is whether uncertainty-aware patchwise physics inference can recover within-leaf spatial heterogeneity, support reliable patch-to-leaf aggregation, and remain robust on external-domain leaves. We build reproducible dense and lesion-aware patch manifests from the 174-leaf primary set and the 18.7 external set, enforce parent-leaf leakage-safe splits, and stage real patch-level oracle/refinement/modeling jobs through resume-safe SLURM pipelines. This draft reports completed data/split/heterogeneity infrastructure and provides the full protocol for heavy optimization/training stages.

1 Full-Leaf vs Patch Distinctness

The prior manuscript studies one parameter vector per full leaf. This new paper changes the unit of inference to patches and introduces: (i) patch-level routing and refinement, (ii) patch-to-leaf aggregation operators, and (iii) within-leaf heterogeneity descriptors. Therefore this is not a revision of the full-leaf manuscript but a separate study design and contribution.

2 Problem Formulation

Given patches I_p extracted from a parent leaf L, we define patch oracle parameters

$$\theta_p^* = \arg\min_{\theta} \mathcal{S}(I_p, \theta), \tag{1}$$

patch prediction $\hat{\theta}_p = f_\phi(I_p)$, uncertainty u_p from ensembles, and routing policy

$$\pi(u_p) \in \{\text{direct}, \text{refine}, \text{fallback}\}. \tag{2}$$

Leaf-level estimates are aggregated from patch outputs via mean, uncertainty-weighted, lesion-weighted, top-k confident, and route-mixture operators.

3 Data and Patch Construction

Two source domains are used: primary `leaves` and external `18.7`. We construct dense and lesion-aware patches at {224,256} with strides {112,128}. Parent-leaf identity is preserved in all manifests to prevent leakage.

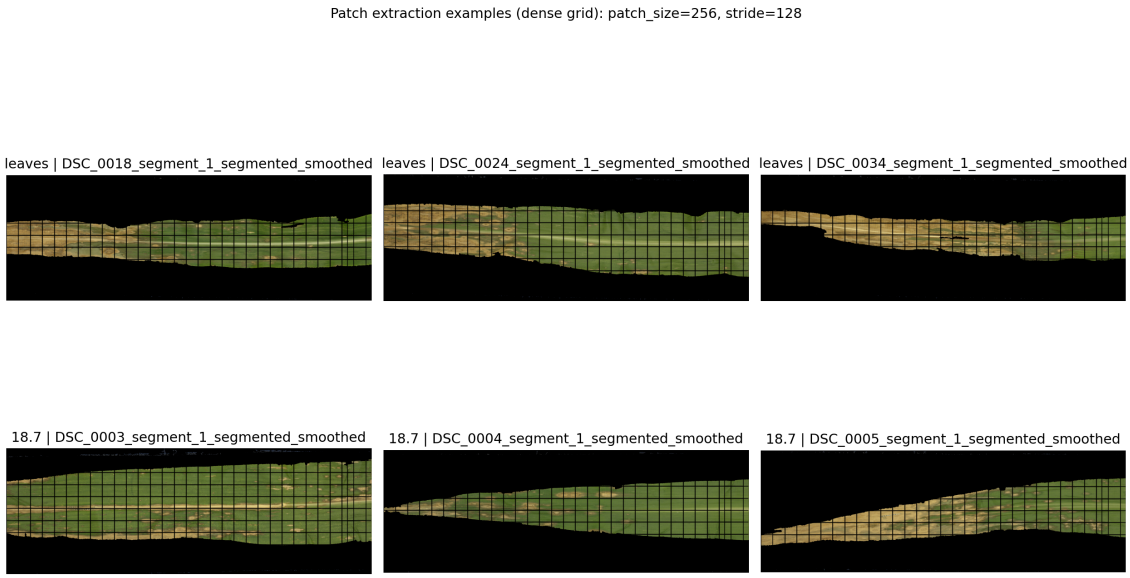

Figure 1: Patch extraction examples (dense-grid overlay) on primary and external leaves.

4 Current Run Artifacts (Completed)

Completed outputs in this run include:

- Patch manifests: `patch_manifest_all.csv`, `patch_manifest_dense.csv`, `patch_manifest_lesionaware.cs`
- Leakage-safe split family: leaf-level, patch-level, pseudo-OOD, domain, parameter-extremes, and learning-curve subsets.
- Initial heterogeneity baselines from lesion-fraction proxies.
- SLURM/tmux resume-safe job scaffolding for heavy stages.

Table 1: Patch counts from current manifests.

Selection	Domain	Num Patches	Num Parent Leaves
dense	leaves	194,704	174
dense	18.7	105,582	78
lesionaware	leaves	101,673	174
lesionaware	18.7	50,994	77

5 Staged Heavy Experiments

The following are configured but intentionally not fabricated in this local turn:

1. Real patch-level oracle optimization at scale.
2. Patch-level model training and uncertainty calibration.

3. Patch-level real short-refinement benchmarking.

4. Full geometry/manifold/hull analysis on learned/refined/oracle patch parameters.

These stages are submitted/resumed via `slurm/submit_all.sh` and logged in `reports/submitted_jobs.txt`.

6 Discussion

Initial patch manifests already show strong domain asymmetry and broad within-leaf lesion-fraction variation, supporting the central patch-study hypothesis: spatially heterogeneous leaves require local inference and selective compute allocation across patches instead of single-vector full-leaf predictions.

7 Conclusion

This draft establishes a distinct patchwise selective semi-amortized study framework with reproducible data/split infrastructure and resume-safe execution for heavy stages. It is a separate paper track from the full-leaf manuscript and targets spatial heterogeneity and patch-to-leaf aggregation as primary contributions.

A Execution Notes

All study artifacts in this draft are written under: `experiments/exp_v1/outputs/selective_semiamortized_p`